SUMMARY OF COST

Bill No.	Description	Amount (Rs.)	Figure in Crore
A	**Civil Works**		
1	SITE CLEARANCE AND DISMANTALING	2,71,49,647	2.71
2	EARTH WORKS	2,16,47,19,927	216.47
3	GRANULAR SUB-BASE AND BASE COURSES	2,46,28,24,986	246.28
4	BITUMINOUS COURSES AND CONCRETE PAVEMENTS	1,27,14,46,928	127.14
5	CROSS DRAINAGE WORKS	2,72,83,23,192	272.83
6	BRIDGES / UNDERPASS	27,61,76,075	27.62
7	REPAIR OF EXISTING STRUCTURES	1,54,320	0.02
8	DRAINAGE AND PROTECTIVE WORKS	5,79,43,89,633	579.44
9	TRAFFIC SIGNS, MARKINGS AND ROAD APPURTENANCES	35,52,64,745	35.53
10	MISCELLANEOUS	16,97,61,090	16.98
	Sub Total for Civil Works (A)	**15,25,02,10,542**	1525.02
B	Utility Shifting Cost	2,50,00,000	2.50
	Sub Total (A+B)	**15,27,52,10,542**	1527.52
C	Physical Contigencies @5% on (A+B)	76,37,60,527	76.38
	Grand Total(A+B+C)	**16,03,89,71,069**	1603.90
	Length of the package Road (km)	**115.5**	**Km**
	Cost per km on TPC	**13.9**	**Crore**

Page Collection

Bill No.	Description			Amount in figure Rs.
1	**Site Clearance and Dismantling**		Page No.3	2,63,12,882
			Page No.4	8,36,765
		Sub Total		**2,71,49,647**
2	**Earthworks**		Page No.5	2,16,47,19,927
		Sub Total		**2,16,47,19,927**
3	**Granular Sub-Base and Base Courses**		Page No.6	2,46,28,24,986
		Sub Total		**2,46,28,24,986**
4	**Bituminous Courses**		Page No.7	1,27,14,46,928
		Sub Total		**1,27,14,46,928**
5	**Cross Drainage Works(Culverts)**		Page No.8	1,50,32,54,169
			Page No.9	1,22,50,69,023
		Sub Total		**2,72,83,23,192**
6	**Bridges/ Underpasses**		Page No.10	18,59,72,011
			Page No.11	9,90,708
			Page No.12	8,01,98,441
			Page No.13	90,14,915
		Sub Total		**27,61,76,075**
7	**Repair of Existing Structures**		Page No.14	1,54,320
		Sub Total		**1,54,320**
8	**Drainage and Protective Works**		Page No.15	2,13,45,94,975
			Page No.16	3,65,97,94,658
		Sub Total		**5,79,43,89,633**
9	**Traffic Signs, Markings and Road Appurtenances**		Page No.17	25,28,78,277
			Page No.18	10,23,86,468
		Sub Total		**35,52,64,745**
10	**Miscellaneous**		Page No.19	16,97,61,090
		Sub Total		**16,97,61,090**

BILL OF QUANTITIES (BOQ)
BILL NO. 1:GENERAL AND SITE CLEARANCE

Item No	Description	Unit	Quantity	Unit Rate	Amount (Rs.)
1.01	Clearing and Grubbing of Roadway as per Technical Specification Clause 201.	Ha	250	55,705	1,39,30,662
1.02	Dismantling of Existing Structures or any Existing Civil Works within Right of Way, as per Technical Specification Clause 202.				
	(i) Brick Masonry	cum	Nil	357	-
	(ii) Stone Masonry	cum	20,905	357	74,63,085
	(iii) Lime/Plain Cement Concrete in all grades	cum	1,140	542	6,17,880
	(iv) Reinforced / Pre-stressed Cement Concrete in all grades	cum	1,310	916	11,99,960
	(v) Bituminous courses	cum	6,005	336	20,17,640
	(vi) Non Bituminous courses	cum	Nil	578	-
	(vii) Cement concrete Pavements	cum	Nil	1,776	-
	(viii) Ordinary KM stone/Guard stone/Sign Post	No.	100	253	25,300
	(ix) Hectometre Stone	No.	Nil	51	-
	(x) 5th KM stone	No.	15	435	6,525
	(xi) Kerb stone	m	Nil	17	-
	(xii) All types of Fencing including posts, foundation concrete, back filling of pit etc. complete.	m	Nil	55	-
	(xiii) RCC Railing	m	Nil	258	-
	(xiv) Guard Rails/W Beam Crash Barrier	m	Nil	87	-
	(xv) Brick Pitching / Brick soling	cum	Nil	229	-
	(xvi) Removal of abandoned Telephone / Electric Poles and Lines including excavation and dismantling of foundation concrete and lines under the supervision of concerned department.	m	Nil	194	-
	(xvii) Removing all types of hume pipes and stacking serviceable material with all leads & lifts including earthwork and dismantling of masonry works.		Nil		-
	(a) Up to 600 mm dia	m	Nil	206	-
	(b) Above 600 mm to 900 mm dia	m	3,770	279	10,51,830
	(c) Above 900 mm dia	m	Nil	478	-
	(xviii) Dismantling stone pitching/ dry stone spalls	cum	Nil	245	-
	(xix) Dismantling of CI Water Pipe Line up to 1000 mm dia.	m	Nil	146	-
	Carried to Page Collection				***2,63,12,882***

BILL OF QUANTITIES
BILL NO. 6: NEW BRIDGES

Item	Description	Unit	Quantity	Unit Rate	Amount (Rupees)
6.10	Bearings, of following Type, as per drawings and Technical Specification Section 2000				
a	Tar Paper Bearings	sqm	53	140	7,420
b	Elastomeric Bearings	cucm	Nil	7	-
c	POT cum PTEE Bearings		Nil		-
i	Fixed Bearings, having Capacity of		Nil		-
	925 T ± 25 T	nos	Nil	1,61,823	-
	525 T ± 25 T	nos	Nil	91,845	-
	325 T ± 25 T	nos	Nil	29,591	-
	275 T ± 25 T	nos	Nil	25,039	-
ii	Free / Guided Bearings, having Capacity of		Nil		-
	925 T ± 25 T	nos	Nil	2,35,298	-
	825 T ± 25 T	nos	Nil	2,09,861	-
	725 T ± 25 T	nos	Nil	1,84,423	-
	675 T ± 25 T	nos	Nil	1,71,704	-
	575 T ± 25 T	nos	Nil	1,46,267	-
	525 T ± 25 T	nos	Nil	1,33,548	-
	325 T ± 25 T	nos	Nil	41,832	-
	275 T ± 25 T	nos	16	35,396	5,66,336
iii	Pin Bearings, having Capacity of		Nil		-
	50 T ± 12.5 T	nos	Nil	16,783	-
	75 T ± 12.5 T	nos	Nil	25,174	-
	100 T ± 12.5 T	nos	Nil	33,565	-
	125 T ± 12.5 T	nos	Nil	50,682	-
	150 T ± 12.5 T	nos	4	71,289	2,85,156
iv	Metallic Guide Bearings, having Capacity of		Nil		-
	25 T ± 12.5 T	nos	Nil	8,237	-
	50 T ± 12.5 T	nos	Nil	16,475	-
	75 T ± 12.5 T	nos	Nil	24,712	-
	100 T ± 12.5 T	nos	4	32,949	1,31,796
6.11	Elastomeric Stopper Pad for Seismic Restrainer as per drawings and Technical Specification 2000	cu.cm	Nil	7	-
	Carried to Page Collection				***9,90,708***

BILL OF QUANTITIES (BOQ)
BILL NO. 8: DRAINAGE AND PROTECTIVE WORKS

Item No.		Unit	Quantity	Unit Rate	Amount (Rs.)
8.01	Lined Random Rubble Masonry Drain on Hill Side, Embankment Toe and Hill Crest Complete as per drawings and Technical Specification Section 300 and 1400 and as directed by the Engineer.	m	2,17,700	4,936	1,07,45,67,200
8.02	Covered RCC Rectangular Drain including Reinforcement complete as per drawing and Technical Specification Sections 300, 1000, 1400,1500,1600, 1700 and as directed by Engineer	m	12,444	20,878	25,98,13,077
8.03	Subsurface Drainage complete as per drawings and Technical Specification Section 309				
a	100 mm diameter PVC Perforated Closed Jointed wrapped with Geo-synthetic as per drawings and Technical Specification	m	780	470	3,66,600
b	Aggregate Drain below Hill Side Drain as per drawings and Technical Specification	m	4,628	417	19,29,876
8.04	RCC Retaining Wall, Gabion Wall				
a	Excavation for Foundation of Structure, as per drawings and Technical Specification, in				
i	All types of soils and Ordinary Rocks as per Technical specification Clause 304.	cum	2,53,976	253	6,42,55,928
ii	Hard Rock (requiring blasting) as per Technical Specification Clause 302 and 304.	cum	Nil	416	-
iii	Hard Rock (using controlled blasting) as per Technical Specification Clause 302 and 304.	cum	Nil	409	-
iv	Hard Rock (blasting prohibited) as per Technical Specification Clause 304	cum	1,08,847	546	5,94,30,462
b	PCC (M 15 Grade) in Levelling Course below Foundation complete as per drawings and Technical Specification Section 1700, 2100, 2500 and 2700.	cum	96	12,866	12,35,136
c	RCC in Substructure for Retaining Wall excluding reinforcement, in following grades, complete as per Drawings and Technical Specification Section 1700 and 2200				
i	M25 Grade	cum	Nil	16,074	-
ii	M 30 Grade	cum	648	16,117	1,04,43,816
d	Steel Reinforcement (Fe 500 D) in Retaining Wall complete as per drawings and Technical Specification Section 1600	MT	66	91,310	60,26,460
e	Weep holes (AC/PVC/HDPE) in Earth Retaining Structures complete as per drawings and Technical Specification Section 2700.	Nos	480	658	3,15,840
f	Back filling behind Earth Retaining Structures Complete as per drawings and Technical Specification Clause 305.	cum	1,64,523	3,980	65,48,01,540
g	Filter Media behind Earth Retaining Structures complete as per Technical Specification Section 2500.	cum	360	3,914	14,09,040
h	Boulder / Concrete Block Pitching on Cut / Embankment Slopescomplete as per drawings and Technical Specification Section 2500.	cum	Nil	4,283	-
	Carried to Page Collection				***2,13,45,94,975***

Project Road : Chakabama Zunheboto Road **Annexure 2.1**

BOQ Item No.

2.01 Project Road Locations not covered under BoQ item 2.02 , 2.03 and 2.04

Cut Quantity	74,19,039	(Summation of Volume under Col. 2 of Annexure 2.1 (c) minus Total Subgrade shape volume under Col. 10)

2.02 Project Road Locations having Sand Stone away by more than 200 m of habitated area

From	To	Cut Quantity	Remarks
8530	8580	547	Summation of respective Volumes under Col. 2 of Annexure 2.1 (c)
69500	70740	34,786	
71300	71400	12,971	
72080	72640	38,804	
73700	73900	8,271	
75600	75700	72	
79580	79900	40,322	
83300	83500	8,251	
83900	84100	18,070	
84200	84390	7,371	
85290	85500	15,966	
88040	88100	1,595	
Total		1,87,027	

2.03 Project Road Locations having Sand Stone within 200 m of habitated area

From	To	Cut Quantity	Remarks
8330	8530	7,983	Summation of respective Volumes under Col. 2 of Annexure 2.1 (c)
75500	75600	0	
83700	83900	6,185	
85120	85250	30,647	
88100	88340	53,676	
Total		98,491	

2.04 Project Road Locations having Sand Stone within 20 m of habitated area

From	To	Cut Quantity	Remarks
8290	8330	355	Summation of respective Volume under Col. 2 of Annexure 2.1 (c)
Total		355	

2.05 Embankment Construction
2,99,969 (Summation of Volumes under Col. 3 of Annexure 2.1 (c))

2.06 Subgrade Preparation from Borrow Earth (CBR > Designed CBR)
1,78,652 (25% of Summation of Volumes under Col. 10 of Annexure 2.1(c))

2.07 Preparation of Subgrade in Cut Formation (Loosening and Compacting to full depth)
5,22,936 (75% of Summation of Volumes under Col. 10 of Annexure 2.1(c))

2.08 Preparation of Subgrade in Rocky Formation
26,040 Surface Area for locations having Sandstone Rock

3.01 GSB 2,96,942 summation of Volumes in Col. 9 in Annexure 2.1 (c)

3.02 WMM 3,22,972 summation of Volumes in Col. 8 in Annexure 2.1 (c)

4.01 Prime Coat 12,84,692 summation of Surface Area in Col. 7 in Annexure 2.1 (c)

4.02 Tack Coat 12,78,010 summation of Surface Area in Col. 5 in Annexure 2.1 (c)

4.03 DBM 64,235 summation of Volumes in Col. 6 in Annexure 2.1 (c)

4.04 BC 38,340 summation of Volumes in Col. 4 in Annexure 2.1 (c)

8.06 Seeding and Mulching
3,97,681 summation of Cutslope Surface Areas for Category II & III stretches as per slope stability surveys

8.07 Geo-synthetic Mat
2,40,791 summation of Cutslope Surface Areas under Landslide prone stretches

8.08 Nailing
1,90,022 summation of Cutslope Surface Areas under Landslide prone stretches

8.09 Temporary Facing
13,853 summation of Surface Areas for first Cut height in Landslide prone stretches

Annexure 2.1 (c): Template Shape / Component Report (from Mx Roads) : CZ Road											Cut Slope			
Station	Cut Volume	Fill Volume	BC Volume	Tack Coat Sur. Area	DBM Volume	Prime Coat Sur. Area	WMM Volume	GSB Volume	Subgrade Volume	<6m Sur. Area	<12 m Sur. Area	> 12 m Sur. Area	RE Wall Sur. Area	R Wall Height
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0														
10	107.4	1.6	3.2	106.7	5.4	108.0	28.1	30.0	65.7					
20	78.4	1.9	3.2	106.7	5.4	108.0	28.1	30.0	65.7					
30	56.5	5.7	3.2	106.7	5.4	108.0	28.1	30.0	65.7					
40	26.5	20.3	3.2	106.7	5.4	108.0	28.1	30.0	65.7					
50	9.7	30.5	3.2	106.7	5.4	108.0	28.1	30.0	65.7					
60	13.9	44.6	3.2	106.7	5.4	108.0	28.1	30.0	65.7					
70	23.6	52.1	3.1	103.3	5.2	104.0	27.1	28.6	56.2					
80	68.7	26.0	3.0	100.0	5.1	102.0	26.0	27.3	58.3					
90	95.7	7.4	3.0	100.0	5.1	102.0	26.0	27.3	58.3					
100	70.2	18.3	3.0	100.0	5.1	102.0	26.0	27.3	58.3					
110	36.3	56.7	3.0	100.0	5.1	102.0	26.0	27.3	58.3					
120	7.2	115.3	3.0	100.0	5.1	102.0	26.0	27.3	58.3					
130	5.1	112.2	3.1	103.3	5.2	104.0	27.0	28.6	61.8					
140	6.5	104.2	3.1	103.3	5.2	104.0	27.0	28.6	56.3					
150	6.2	111.2	3.0	100.0	5.1	102.0	26.0	27.3	58.3					
160	28.6	65.0	3.0	100.0	5.1	102.0	26.0	27.3	58.3					
169	66.0	22.3	2.7	90.0	4.6	92.0	23.6	24.8	53.0	3.0				
170	9.3	1.1	0.3	10.0	0.5	10.0	2.4	2.5	5.3	1.0				
180	118.8	5.6	3.1	103.3	5.2	104.0	26.6	27.9	59.5	5.0				
184	51.8	0.3	1.3	43.3	2.2	44.0	11.2	11.7	25.0	1.0				
190	75.3	0.4	1.9	63.3	3.2	64.0	16.6	17.4	36.9	3.0				
198	106.2	0.2	2.5	83.3	4.3	86.0	22.0	22.9	48.7	4.0				
200	35.8	0.1	0.8	26.7	1.4	28.0	7.1	7.4	15.8	1.0				
210	141.0	6.5	3.5	116.7	5.8	116.0	29.7	31.0	65.8	3.0				
220	115.7	17.6	3.5	116.7	5.8	116.0	29.8	31.0	65.8					
230	116.4	24.1	3.5	116.7	5.8	116.0	29.8	31.0	65.8					
240	129.9	19.3	3.6	120.0	6.0	120.0	30.9	32.5	69.9					
250	109.3	6.8	3.6	120.0	6.1	122.0	31.8	33.9	73.9					
251	11.7		0.5	16.7	0.8	16.0	3.9	4.2	9.2					
260	77.4	4.3	3.1	103.3	5.2	104.0	27.1	28.9	63.0	4.0				
265	38.2	5.8	1.6	53.3	2.7	54.0	14.3	15.2	33.1	1.0				
270	40.5	8.6	1.8	60.0	3.0	60.0	15.5	16.5	36.0	1.0				
280	68.1	15.4	3.2	106.7	5.4	108.0	27.9	29.8	65.1	7.0				
280	1.6	0.3	0.1	3.3	0.1	2.0	0.7	0.8	1.7					
290	84.8	7.4	3.2	106.7	5.4	108.0	28.1	30.0	65.7	9.0				
300	98.0	1.0	3.2	106.7	5.4	108.0	28.1	30.0	65.7	5.0				
300	4.7	0.1	0.2	6.7	0.3	6.0	1.4	1.5	3.2					
310	79.3	11.8	3.0	100.0	5.1	102.0	26.7	28.5	62.6					
320	88.6	13.5	3.2	106.7	5.4	108.0	28.1	30.0	65.7					
330	129.0	3.1	3.1	103.3	5.2	104.0	26.8	28.3	55.6					
330	1.5				0.1	2.0	0.3	0.3	0.6					
340	180.0	1.5	3.1	103.3	5.2	104.0	26.6	27.8	59.4	1.0				
345	109.1		1.6	53.3	2.6	52.0	13.5	14.1	30.0	1.0				
345	11.9		0.2	6.7	0.3	6.0	1.4	1.5	3.1					
350	118.4	0.1	1.5	50.0	2.5	50.0	12.9	13.5	28.7					
360	292.2	0.4	3.3	110.0	5.5	110.0	28.3	29.5	62.8					
366	167.6	0.2	2.1	70.0	3.5	70.0	17.9	18.7	39.7	3.0				
367	11.0		0.2	6.7	0.3	6.0	1.4	1.5	3.1					
370	70.4		1.0	33.3	1.7	34.0	8.7	9.1	19.3	3.0				
380	256.5		3.1	103.3	5.2	104.0	26.7	28.0	59.7	7.0				
382	48.5		0.6	20.0	0.9	18.0	4.8	5.1	10.8					
390	167.5	4.1	2.5	83.3	4.3	86.0	22.0	23.3	50.5					
400	116.9	22.1	3.2	106.7	5.4	108.0	28.1	30.0	65.7					
410	71.6	18.1	3.3	110.0	5.5	110.0	28.8	30.7	67.1					
411	8.3	0.2	0.3	10.0	0.6	12.0	3.0	3.2	6.9					
420	86.6	7.2	3.0	100.0	5.0	100.0	26.0	27.4	53.9	9.0				
430	207.2	13.3	3.3	110.0	5.5	110.0	28.3	29.5	62.8		38.0			
440	286.0	15.9	3.3	110.0	5.5	110.0	28.3	29.5	62.8		110.0			
450	216.0	22.2	3.3	110.0	5.5	110.0	28.3	29.5	62.8	17.0				
460	210.1	22.1	3.3	110.0	5.5	110.0	28.3	29.5	62.8	36.0				
470	356.5	18.4	3.3	110.0	5.5	110.0	28.3	29.5	62.8	45.0				
480	548.2	23.0	3.3	110.0	5.5	110.0	28.3	29.5	62.8	59.0				
490	816.9	14.7	3.3	110.0	5.5	110.0	28.3	29.5	62.8		48.0			
500	977.6		3.3	110.0	5.5	110.0	28.3	29.5	62.8		138.0			
510	785.5	19.7	3.3	110.0	5.5	110.0	28.3	29.5	62.8		180.0			
520	327.6	34.3	3.3	110.0	5.5	110.0	28.3	29.5	62.8	5.0				
530	48.7	24.5	3.3	110.0	5.5	110.0	28.3	29.5	62.8					
540	89.2	19.8	3.3	110.0	5.5	110.0	28.3	29.5	62.8	2.0				
550	161.1	17.6	3.3	110.0	5.5	110.0	28.3	29.5	62.8	8.0				
560	185.3	21.7	3.3	110.0	5.5	110.0	28.3	29.5	62.8	13.0				
570	127.0	44.5	3.3	110.0	5.5	110.0	28.3	29.5	62.8					
580	68.8	50.1	3.3	110.0	5.5	110.0	28.3	29.5	62.8					
590	55.7	27.8	3.4	113.3	5.7	114.0	29.3	30.9	66.6					
600	49.9	14.4	3.5	116.7	5.8	116.0	30.3	32.3	70.2					
610	44.1	10.3	3.5	116.7	5.8	116.0	30.3	32.3	70.2					
620	43.1	13.3	3.5	116.7	5.8	116.0	30.3	32.3	70.2					
630	44.7	23.9	3.4	113.3	5.7	114.0	29.3	30.9	60.7					
640	43.6	35.5	3.3	110.0	5.5	110.0	28.3	29.5	62.8					
645	21.6	22.5	1.7	56.7	2.9	58.0	14.7	15.4	32.7					
645	0.4	0.4			0.1	2.0	0.3	0.3	0.6					
645														
650	18.1	20.5	1.5	50.0	2.6	52.0	13.3	13.9	29.5					
660	59.2	30.1	3.3	110.0	5.5	110.0	28.3	29.5	62.8					
670	85.3	18.3	3.3	110.0	5.5	110.0	28.3	29.5	62.8	3.0				
680	85.0	21.9	3.3	110.0	5.5	110.0	28.3	29.5	62.8	5.0				
690	80.0	21.1	3.3	110.0	5.5	110.0	28.3	29.5	62.8	4.0				
700	82.8	11.5	3.3	110.0	5.5	110.0	28.3	29.5	62.8	5.0				
710	94.7	9.0	3.3	110.0	5.5	110.0	28.3	29.5	62.8	6.0				
719	86.5	7.2	2.9	96.7	4.9	98.0	25.3	26.4	56.1	2.0				
720	10.5	0.2	0.4	13.3	0.6	12.0	3.2	3.4	7.2					
730	137.6	3.6	3.3	110.0	5.6	112.0	28.8	30.1	63.9	23.0				
740	230.9	10.2	3.4	113.3	5.7	114.0	29.0	30.3	64.3	49.0				
750	309.6	14.6	3.4	113.3	5.7	114.0	29.0	30.3	64.3	60.0				
760	272.1	11.7	3.4	113.3	5.7	114.0	29.0	30.3	64.3	64.0				
770	140.9	3.8	3.4	113.3	5.7	114.0	29.0	30.3	64.3	31.0				
778	63.8	5.6	2.8	93.3	4.7	94.0	23.9	25.0	53.1	3.0				
780	12.6	2.1	0.5	16.7	0.9	18.0	4.7	4.9	10.4	1.0				
790	71.7	15.9	3.2	106.7	5.4	108.0	27.8	29.1	61.9	4.0				
800	64.6	19.6	3.2	106.7	5.4	108.0	27.5	28.8	61.3	1.0				
810	63.6	16.1	3.2	106.7	5.4	108.0	27.6	28.9	61.5	1.0				
813	19.7	3.2	1.0	33.3	1.6	32.0	8.2	8.6	18.2					
820	48.1	4.0	2.3	76.7	3.9	78.0	19.8	20.7	44.0					
830	69.4	4.1	3.3	110.0	5.5	110.0	28.3	29.5	62.8	1.0				
840	88.4	3.9	3.3	110.0	5.5	110.0	28.3	29.5	62.8	4.0				
850	88.6	7.9	3.3	110.0	5.6	112.0	28.7	30.0	63.7					
855	34.7	5.6	1.5	50.0	2.6	52.0	13.3	13.9	29.5					
860	46.6	5.0	1.9	63.3	3.2	64.0	16.3	17.2	36.9					
870	81.1	19.1	3.6	120.0	6.1	122.0	31.4	33.3	72.4					
880	154.2	24.6	3.6	120.0	6.0	120.0	30.8	32.4	63.7	2.0				
890	317.8	9.4	3.5	116.7	5.8	116.0	29.8	31.0	65.8	17.0				
900	429.8		3.5	116.7	5.8	116.0	29.8	31.0	65.8	34.0				

QUANTITY CALCULATION FOR 1X1.2 DIA PIPES

Sl. No.	Proposed Chainage (km)	Culvert Designation	Pipe Length	Excavation	Backfilling	Conc Foundation	Conc Substructure	PCC M15 Levelling Coarse	Filter Media	Painting	Lettering
1	75.220	1*1.2	12.500	137.41	137.21	12.96	20.40	3.47	30.24	2.28	50
2	75.120	1*1.2	10.000	137.14	137.21	12.96	20.40	3.47	30.24	2.28	50
3	94.460	1*1.2	12.500	165.03	179.18	14.58	24.00	3.88	33.84	2.28	50
4	90.990	1*1.2	12.500	124.80	137.21	12.96	20.40	3.47	30.24	2.28	50
5	96.090	1*1.2	12.500	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50
6	91.150	1*1.2	12.500	109.88	137.21	12.96	20.40	3.47	30.24	2.28	50
7	35.020	1*1.2	12.500	165.49	179.18	14.58	24.00	3.88	33.84	2.28	50
8	82.860	1*1.2	10.000	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50
9	81.280	1*1.2	12.500	130.57	137.21	12.96	20.40	3.47	30.24	2.28	50
10	93.220	1*1.2	10.000	106.96	137.21	12.96	20.40	3.47	30.24	2.28	50
11	97.260	1*1.2	12.500	126.55	137.21	12.96	20.40	3.47	30.24	2.28	50
12	89.310	1*1.2	12.500	110.25	137.21	12.96	20.40	3.47	30.24	2.28	50
13	88.240	1*1.2	12.500	110.02	137.21	12.96	20.40	3.47	30.24	2.28	50
14	76.250	1*1.2	10.000	120.19	137.21	12.96	20.40	3.47	30.24	2.28	50
15	14.935	1*1.2	12.500	120.65	137.21	12.96	20.40	3.47	30.24	2.28	50
16	75.985	1*1.2	12.500	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50
17	34.970	1*1.2	10.000	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50
18	95.960	1*1.2	12.500	165.21	179.18	14.58	24.00	3.88	33.84	2.28	50
19	50.170	1*1.2	10.000	126.97	137.21	12.96	20.40	3.47	30.24	2.28	50
20	88.880	1*1.2	10.000	103.57	137.21	12.96	20.40	3.47	30.24	2.28	50
21	80.080	1*1.2	12.500	98.40	179.18	14.58	24.00	3.88	33.84	2.28	50
22	4.430	1*1.2	10.000	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50
23	76.420	1*1.2	10.000	98.40	179.18	14.58	24.00	3.88	33.84	2.28	50

Sl. No.	Proposed Chainage (km)	Culvert Designation	Pipe Length	Excavation	Backfilling	Conc Foundation	Conc Substructure	PCC M15 Levelling Coarse	Filter Media	Painting	Lettering
249	31.360	1*1.2	12.500	98.42	137.21	12.96	20.40	3.47	30.24	2.28	50
250	62.575	1*1.2	12.500	91.97	137.21	12.96	20.40	3.47	30.24	2.28	50
251	66.250	1*1.2	12.500	114.86	137.21	12.96	20.40	3.47	30.24	2.28	50
252	66.630	1*1.2	10.000	104.35	137.21	12.96	20.40	3.47	30.24	2.28	50
253	56.905	1*1.2	10.000	103.13	137.21	12.96	20.40	3.47	30.24	2.28	50
254	78.570	1*1.2	12.500	103.47	137.21	12.96	20.40	3.47	30.24	2.28	50
255	99.160	1*1.2	12.500	107.30	137.21	12.96	20.40	3.47	30.24	2.28	50
256	70.780	1*1.2	12.500	93.83	137.21	12.96	20.40	3.47	30.24	2.28	50
257	0.110	1*1.2	10.000	100.54	137.21	12.96	20.40	3.47	30.24	2.28	50
258	23.370	1*1.2	12.500	100.66	137.21	12.96	20.40	3.47	30.24	2.28	50
259	33.563	1*1.2	12.500	100.25	137.21	12.96	20.40	3.47	30.24	2.28	50
260	100.460	1*1.2	12.500	99.57	137.21	12.96	20.40	3.47	30.24	2.28	50
261	20.345	1*1.2	12.500	91.43	276.06	21.03	33.42	4.68	41.04	2.28	50
262	31.730	1*1.2	12.500	96.35	137.21	12.96	20.40	3.47	30.24	2.28	50
263	75.625	1*1.2	12.500	64.78	137.21	12.96	20.40	3.47	30.24	2.28	50
264	98.730	1*1.2	12.500	103.57	137.21	12.96	20.40	3.47	30.24	2.28	50
265	99.460	1*1.2	12.500	75.41	137.21	12.96	20.40	3.47	30.24	2.28	50
266	99.350	1*1.2	12.500	59.70	179.18	14.58	24.00	3.88	33.84	2.28	50
267	17.360	1*1.2	10.000	73.47	296.31	21.33	35.46	4.90	44.64	2.28	50
		Total	**3167.500**	**30644.178**	**38742.529**	**3557.565**	**5635.770**	**946.523**	**8254.080**	**608.760**	**13350.000**

QUANTITY CALCULATION FOR 2X1.2 DIA PIPES

268	50.870	2*1.2	25.000	107.50	137.21	18.150	21.74	3.596	30.24	2.28	50
269	51.770	2*1.2	25.000	112.49	137.21	18.150	21.74	3.596	30.24	2.28	50
270	56.415	2*1.2	25.000	131.68	137.21	18.150	21.74	3.596	30.24	2.28	50

QUANTITY CALCULATION FOR 2.0X2.0X2.0 SIZED BOX CULVERTS

Sl.No.	Proposed Chainage	Designation	Excavation	RCC for box	Approach slab RCC	Exp. Gap	Lettering	Drainage Spout	Painting	Levelling course	Excavation For Return Wall	RCC in Return WALL	Filter Media	Backfill	RCC for Counterfort	RCC for Wall
	(km)		m^3	m^3	m^3	lm	nos.	nos.	m^2	m^3	m^3	m^3	m^3	m^3	m^3	m^3
1	87.270	1.22.00	109.76	56.51	26.71	21.80	50.00	4.00	9.88	15.87	80.08	22.24	20.16	91.48	0.00	0.00
2	92.665	1.22.00	109.76	56.51	26.71	21.80	50.00	4.00	9.88	15.87	80.08	22.24	20.16	91.48	0.00	0.00
3	27.670	1.22.00	109.76	56.51	26.71	21.80	50.00	4.00	9.88	15.87	80.08	22.24	20.16	91.48	0.00	0.00
4	36.405	1.22.00	100.70	52.19	24.50	20.00	50.00	4.00	9.88	14.59	80.08	22.24	20.16	91.48	0.00	0.00
5	13.520	1.22.00	112.78	57.95	27.44	22.40	50.00	4.00	9.88	16.30	80.08	22.24	20.16	91.48	0.00	0.00
6	71.230	1.22.00	106.74	55.07	25.97	21.20	50.00	4.00	9.88	15.44	80.08	22.24	20.16	91.48	0.00	0.00
7	0.890	1.22.00	115.81	59.39	28.18	23.00	50.00	4.00	9.88	16.73	80.08	22.24	20.16	91.48	0.00	0.00
8	94.160	1.22.00	100.70	52.19	24.50	20.00	50.00	4.00	9.88	14.59	80.08	22.24	20.16	91.48	0.00	0.00
9	2.275	1.22.00	115.81	59.39	28.18	23.00	50.00	4.00	9.88	16.73	80.08	22.24	20.16	91.48	0.00	0.00
10	26.550	1.22.00	115.80	59.39	28.18	23.00	50.00	4.00	9.88	16.73	80.08	22.24	20.16	91.48	0.00	0.00
11	16.490	1.22.00	109.76	56.51	26.71	21.80	50.00	4.00	9.88	15.87	80.08	22.24	20.16	91.48	0.00	0.00
12	80.960	1.22.00	106.74	55.07	25.97	21.20	50.00	4.00	9.88	15.44	80.08	22.24	20.16	91.48	0.00	0.00
13	19.245	1.22.00	106.74	55.07	25.97	21.20	50.00	4.00	9.88	15.44	80.08	22.24	20.16	91.48	0.00	0.00
14	2.570	1.22.00	100.70	52.19	24.50	20.00	50.00	4.00	9.88	14.59	80.08	22.24	20.16	91.48	0.00	0.00
15	111.760	1.22.00	100.70	52.19	24.50	20.00	50.00	4.00	9.88	14.59	80.08	22.24	20.16	91.48	0.00	0.00
16	9.160	1.22.00	112.78	57.95	27.44	22.40	50.00	4.00	9.88	16.30	80.08	22.24	20.16	91.48	0.00	0.00
17	25.808	1.22.00	112.78	57.95	27.44	22.40	50.00	4.00	9.88	16.30	80.08	22.24	20.16	91.48	0.00	0.00
18	55.210	1.22.00	112.78	57.95	27.44	22.40	50.00	4.00	9.88	16.30	80.08	22.24	20.16	91.48	0.00	0.00
19	107.560	1.22.00	100.70	52.19	24.50	20.00	50.00	4.00	9.88	14.59	80.08	22.24	20.16	91.48	0.00	0.00

Sl.No.	Proposed Chainage	Designation	Excavation	RCC for box	Approach slab RCC	Exp. Gap	Lettering	Drainage Spout	Painting	Levelling course	Excavation For Return Wall	RCC in Return WALL	Filter Media	Backfill	RCC for Counterfort	RCC for Wall
	(km)		m^3	m^3	m^3	lm	nos.	nos.	m^2	m^3	m^3	m^3	m^3	m^3	m^3	m^3
220	3.730	1.22.00	56.05	52.19	24.50	20.00	50.00	4.00	9.88	14.59	69.55	29.20	24.96	147.43	0.77	5.56
221	71.180	1.22.00	12.08	55.07	25.97	21.20	50.00	4.00	9.88	15.44	45.00	37.86	29.76	197.54	1.40	10.08
222	70.430	1.22.00	12.77	57.95	27.44	22.40	50.00	4.00	9.88	16.30	45.00	37.86	29.76	197.54	1.46	10.48
		Total	17628.35	12398.75	5853.54	4778.40	11100.00	888.00	2193.36	3480.05	14641.95	5024.20	4533.12	20967.40	10.443115	75.19043

S. No.	Proposed Chainage	Designation	Excavation	RCC for box	Approach slab RCC	Exp. Gap	Lettering	Drainage Spout	painting	levelling course	Excavation For Return Wall	RCC for Return Wall	Filter Media	Backfill	RCC for Counter fort	RCC for Wall
	(km)		m^3	m^3	m^3	lm	nos.	nos.	m^2	m^3	m^3	m^3	m^3	m^3	m^3	m^3
114	87.770	1.33.00	131.47	97.16	27.44	22.40	50.00	4.00	14.59	18.63	96.30	37.86	29.76	197.54	0.50	5.64
115	17.950	1.33.00	117.38	87.40	24.50	20.00	50.00	4.00	14.59	16.67	96.30	37.86	29.76	197.54	0.65	7.37
116	97.940	1.33.00	45.11	87.40	24.50	20.00	50.00	4.00	14.59	16.67	59.17	37.86	29.76	197.54	0.68	7.71
117	85.990	1.33.00	97.52	87.40	24.50	20.00	50.00	4.00	14.59	16.67	86.10	37.86	29.76	197.54	0.90	10.17
118	43.605	1.33.00	20.15	99.60	28.18	23.00	50.00	4.00	14.59	19.12	67.80	67.09	39.36	374.56	1.73	19.49
	Total		22549.65	11800.47	3173.58	2590.68	5900.00	472.00	1671.70	2184.30	12594.60	3535.45	2988.48	17924.64	6.26	72.74

SL. No.	Proposed Chainage (km)	DOWNSTREAM PROTECTION WORKS Excavation	PCC M15	RCC Total	Boulders in Wire Crates	Reinforcement	UPSTREAM PROTECTION WORKS Excavation	Excavation for Catchpit	Excavation Total	PCC M15	RCC	RCC For Catchpit	RCC Total	Boulders in Wire Crates	Reinforcement	Reinforcement for Catchpit	Reinforcement Total
150	74.010	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.73	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
151	69.100	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.73	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
152	32.807	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.73	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
153	74.310	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.73	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
154	35.260	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.73	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
155	8.290	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.73	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
156	26.660	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.73	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
157	3.430	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.73	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
158	45.520	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.73	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
159	48.420	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.73	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
160	89.770	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
161	38.570	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
162	39.150	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
163	32.647	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
164	80.685	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
165	45.355	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
166	44.740	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
167	26.125	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
168	15.635	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
169	28.625	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
170	100.580	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
171	56.635	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
172	34.590	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
173	63.065	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
174	0.310	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
175	95.510	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
176	69.410	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
177	5.210	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
178	30.620	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
179	55.360	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
180	10.495	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
181	100.280	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
182	39.260	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
183	41.480	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
184	3.290	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
185	98.950	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
186	3.010	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20
187	3.850	43.17	2.58	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20

SL. No.	Proposed Chainage	DOWNSTREAM PROTECTION WORKS							UPSTREAM PROTECTION WORKS										
		Excavation	PCC M15	RCC for flooring+wall +block	Rcc in Front and Back Curtain Wall	RCC Total	Boulders in Wire Crates	Reinforcement	Excavation	Excavation for Catchpit	Excavation Total	PCC M15	RCC	RCC For Catchpit	RCC Total	Boulders in Wire Crates	Reinforcement	Reinforcement for Catchpit	Reinforcement Total
	(km)	m^3	m^3	m^3	m^3	m^3	m^3	MT	m^3	m^3	m^3	m^3	m^3	m^3	m^3	m^3	MT	MT	MT
20	55.450	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
21	0.385	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
21	29.313	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
22	111.930	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
22	6.370	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
23	79.550	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
23	48.885	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
24	35.640	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
24	79.000	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
25	47.540	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
25	89.460	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
26	107.100	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
26	5.575	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
27	11.975	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
27	36.100	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
28	86.590	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
28	84.800	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
29	102.470	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
29	15.175	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
30	0.220	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
30	112.050	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
31	107.980	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
31	62.895	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
32	108.300	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
32	78.105	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
33	110.450	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
33	68.540	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
34	104.650	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
34	6.145	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
35	103.980	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
35	42.245	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
36	47.740	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
36	72.000	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
37	73.580	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
37	67.845	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
38	108.560	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
38	37.950	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
39	53.970	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
39	68.680	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39
40	49.700	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39

Detailed Calculation For Upstream Protection Works for 2mx2m Single cell Box Culvert

ITEM NO.	ITEM	DESCRIPTION	UNIT	NO.	L (m)	B (m)	AREA (m^2)	H (m) / Thick. (m)	Volume/Qty (m^3)	SUB HEAD
	Segment 1									
5.01	**Earthwork in excavations:**									
	For RCC Rigid flooring (M-30)		m^3	1.00	3.00	4.40	13.20	0.60	7.92	7.92
	For Hand woven wire crates in flexible apron for protection work in upstream		m^3	1.00	3.00	5.40	16.20	1.30	21.06	21.06
	For curtain wall in segment 1		m^3	1.00	5.40	1.70		1.40	12.85	12.85
									Sub Total	41.83
	70% in Ordinary Soil									29.28
	30% in Hard Rock									12.56
5.02	PCC M-15 below curtain wall	100 mm thick.	m^3	1.00	5.40	1.70		0.10	0.92	0.92
5.02	PCC M-15 below RCC flooring	100 mm thick.	m^3	1.00	3.00	4.40	13.20	0.10	1.32	1.32
									Sub Total	2.24
5.04	RCC rigid flooring M-30			1.00	3.00	4.40	13.20	0.20	2.64	2.64
5.04	RCC Side walls over rigid flooring M-30	0.2 m thick.	m^3	2.00	3.16	0.20		1.00	1.26	1.26
									Sub Total	3.90
5.04	Rcc Curtain wall 1		m^3	1.00	5.40		0.84		4.54	4.54
5.20	Hand woven wire crates in flexible apron for	1000 mm thick.	m^3	1.00	3.00	5.40	16.20	1.00	16.20	16.20
5.07	Reinforcement in Rcc curtain wall		t	1.00	4.54	@50kg/m3 of concrete			226.80	0.227
5.07	Reinforcement in RCC flooring		t	1.00	3.90	@50kg/m3 of concrete			195.25	0.195
	Segment 2									
5.01	**Earthwork in excavations:**									
	Rcc slope protection structure in segment 2		m^3	1.00	1.00	3.40		0.60	2.04	2.04
	For curtain wall in segment 2		m^3	1.00	3.40	0.40		1.50	2.04	2.04
									Sub Total	2.04
	50% in Ordinary Soil									1.02
	50% in Hard rock									1.02
5.02	Pcc below Curtain wall			1.00	3.40	0.40		0.10	0.14	0.14
5.02	Pcc below Concrete for slope protection	(per meter)	m^3	1.00	1.00	3.40		0.10	0.34	0.34
									Sub Total	0.48
5.04	Rcc Concrete for slope protection structure	(per meter)	m^3	1.00	1.00	3.40		0.20	0.68	0.68
	RCC Side walls over rigid flooring M-30	(per meter)	m^3	2.00	1.00	0.20		1.00	0.40	0.40
5.04	Rcc Curtain wall		m^3	1.00	3.40	0.3		1.50	1.53	1.53
5.04	Buffle pier		m^3	28.00	0.20	0.20		0.20	0.224	0.224
									Sub Total	2.83
5.07	Reinforcement in Rcc curtain wall		t	1.00	1.5300	@50kg/m3 of concrete			76.50	0.077
5.07	Reinforcement in RCC slope protection		t	1.00	1.30	@50kg/m3 of concrete			65.20	0.065
	Segment 3									
5.01	**Earthwork in excavations:**									
	For catchpit		m^3	1.00	1.95	3.80	7.41	1.30	9.63	9.63
	50% in Ordinary Soil									4.82
	50% in Hard Rock									4.82
5.02	PCC M-15 below Catchpit		m^3	1.00	1.85	3.70	6.85	0.10	0.68	0.68
5.04	Rcc M-30 Catchpit walls	A1 (per meter)	m^3	2.00	1.50	0.20		1.00	0.60	0.60
		A2 (per meter)	m^3	1.00	3.40	0.20		1.00	0.68	0.68
									Sub Total	1.28
5.04	RCC M-30 Catchpit base		m^3	1.00	1.85	3.90		0.20	1.44	1.44
5.07	Reinforcement in Rcc catchpit walls		t	1.00	1.28	@50kg/m3 of concrete			64.00	0.064
	Reinforcement in Rcc catchpit base		t	1.00	1.44	@50kg/m3 of concrete			72.15	0.072

Detailed Calculation For Download Protection Works for 2x2m Single Cell Box Culvert

ITEM NO.	ITEM	DESCRIPTION	UNIT	NO.	L (m)	B (m)	AREA (m^2)	H (m) / Thick. (m)	Volume/Qty (m^3)	SUB HEAD TOTAL
5.1	**Earthwork in excavations for protection works**									
	For RCC Rigid flooring (M-30)		m^3	1.00	5.00	4.40	22.00	0.30	6.60	6.60
	For Hand woven wire crates in flexible apron for protection work		m^3	1.00	6.00	5.40	32.40	1.00	32.40	32.40
	For front curtain wall		m^3	1.00	5.40	1.70		1.30	11.93	11.93
	For back curtain wall		m^3	0.00	3.40	0.30		1.00	0.00	0.00
										50.93
	70% in soil									35.65
	30% in ordinary rock									15.28
	Outlet/downstream Protection Work									
5.04	RCC channel/flooring M-30	0.2m thick	m^3	1.00	5.00	4.40		0.20	4.40	4.40
5.04	RCC Side wall of Rigid Flooring M-30			2.00	5.10	0.20		1.00	2.04	2.04
5.04	RCC concrete block		m^3	30.00	0.30	0.20		0.30	0.54	0.54
		Sub Total								**6.98**
5.04	RCC In Front Curtain wall		m^3	1.00	5.40		0.81		4.37	4.37
5.04	RCC in Back Curtain wall		m^3	1.00	3.40	0.30		1.00	1.02	1.02
		Sub Total								**5.39**
5.02	PCC M-15 below RCC Rigid flooring	0.1m thick	m^3	1.00	5.00	4.40		0.10	2.20	2.20
5.02	PCC M-15 below fron Curtain wall		m^3	1.00	5.40	1.70		0.10	0.92	0.92
5.20	Hand woven wire crates in flexible apron for protection work	1m thick	m^3	1.00	6.00	5.40		1.00	32.40	32.40
5.07	Reinforcement in RCC flooring		Mt	1.00	6.98	.@50kg/m3 of concrete			348.98	0.349
5.07	Reinforcement in curtain wall		Mt	1.00	5.39	.@50kg/m3 of concrete			269.70	0.270

Detailed Calculation For Downstream Protection Works for 2mx2m Double cell Box Culvert

ITEM NO.	ITEM	DESCRIPTION	UNIT	NO.	L (m)	B (m)	AREA (m^2)	H (m) / Thick. (m)	Volume/Qty (m^3)	SUB HEAD TOTAL
5.1	**Earthwork in excavations for protection works**									
	For RCC Rigid flooring (M-30)		m^3	1.00	5.00	6.70	33.50	0.30	10.05	10.05
	For Hand woven wire crates in flexible apron for protection work		m^3	1.00	6.00	7.70	46.20	1.00	46.20	46.20
	For front curtain wall		m^3	1.00	7.70	1.70		1.30	17.02	17.02
	For back curtain wall		m^3	0.00	5.70	0.30		1.00	0.00	0.00
										73.27
	70% in soil									51.29
	30% in ordinary rock									21.98
	Outlet/downstream Protection Work									
5.04	RCC channel/flooring M-30	0.2m thick	m^3	1.00	5.00	6.70		0.20	6.70	6.70
5.04	RCC Side wall of Rigid Flooring M-30			2.00	5.10	0.20		1.00	2.04	2.04
5.04	RCC concrete block		m^3	48.00	0.30	0.20		0.30	0.86	0.86
		Sub Total								**9.60**
5.04	RCC in Front Curtain wall		m^3	1.00	7.70		0.81		6.24	6.24
5.04	RCC in Back Curtain wall		m^3	1.00	5.70	0.30		1.00	1.71	1.71
		Sub Total								**7.95**
5.02	PCC M-15 below RCC Rigid flooring	0.1m thick	m^3	1.00	5.00	6.70		0.10	3.35	3.35
5.02	PCC M-15 below fron Curtain wall		m^3	1.00	7.70	1.70		0.10	1.31	1.31
5.20	Hand woven wire crates in flexible apron for protection work	1m thick	m^3	1.00	6.00	7.70		1.00	46.20	46.20
5.07	Reinforcement in RCC flooring		Mt	1.00	9.60	@50kg/m3 of concrete			480.18	0.480
5.07	Reinforcement in curtain wall		Mt	1.00	7.95	.@50kg/m3 of concrete			397.35	0.397

Detailed Calculation For Downstream Protection Works for 3mx3m Double cell Box Culvert

ITEM NO.	ITEM	DESCRIPTION	UNIT	NO.	L (m)	B (m)	AREA (m^2)	H (m) / Thick. (m)	Volume/Qty (m^3)	SUB HEAD TOTAL
5.1	**Earthwork in excavations for protection works**									
	For RCC Rigid flooring (M-30)		m^3	1.00	5.00	8.70	43.50	0.30	13.05	13.05
	For Hand woven wire crates in flexible apron for protection work		m^3	1.00	6.00	9.70	58.20	1.00	58.20	58.20
	For front curtain wall		m^3	1.00	9.70	1.70		1.30	21.44	21.44
	For back curtain wall		m^3	0.00	7.70	0.30		1.00	0.00	0.00
										92.69
	70% in soil									64.88
	30% in ordinary rock									27.81
	Outlet/downstream Protection Work									
5.04	RCC channel/flooring M-30	0.2m thick	m^3	1.00	5.00	8.70		0.20	8.70	8.70
5.04	RCC Side wall of Rigid Flooring M-30			2.00	5.10	0.20		1.00	2.04	2.04
5.04	RCC concrete block		m^3	60.00	0.30	0.20		0.30	1.08	1.08
		Sub Total								**11.82**
5.04	RCC in Front Curtain wall		m^3	1.00	9.70		0.81		7.86	7.86
5.04	RCC in Back Curtain wall		m^3	1.00	7.70	0.30		1.00	2.31	2.31
		Sub Total								**10.17**
5.02	PCC M-15 below RCC Rigid flooring	0.1m thick	m^3	1.00	5.00	8.70		0.10	4.35	4.35
5.02	PCC M-15 below fron Curtain wall		m^3	1.00	9.70	1.70		0.10	1.65	1.65
5.20	Hand woven wire crates in flexible apron for protection work	1m thick	m^3	1.00	6.00	9.70		1.00	58.20	58.20
5.07	Reinforcement in RCC flooring		Mt	1.00	11.82	.@50kg/m3 of concrete			590.98	0.591
5.07	Reinforcement in curtain wall		Mt	1.00	10.17	.@50kg/m3 of concrete			508.35	0.508

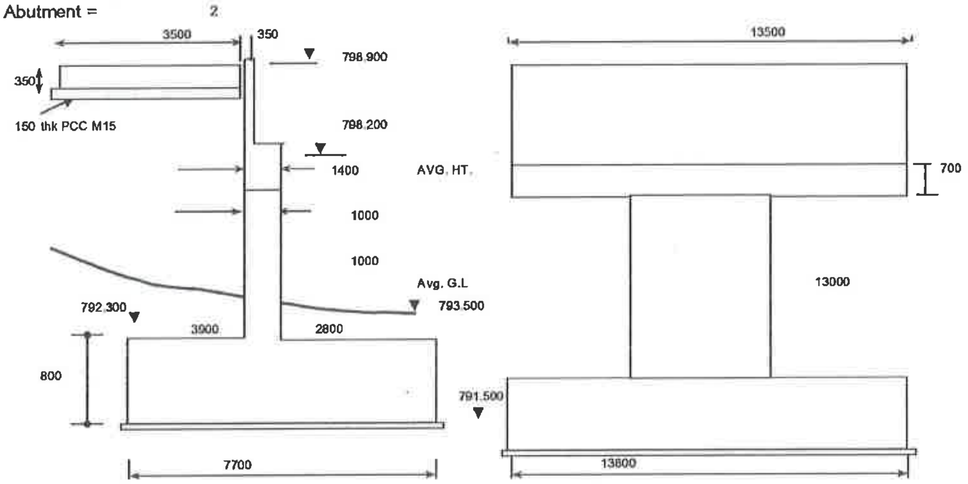
Summary Of Quantities for bridges of CZ road

Item No. as per Quantity Estimates	Item No. as per BOQ	Description	Unit	RCC Slab at KM 7.490	RCC voided Slab at KM. 8.690	RCC voided Slab at KM. 13.455	RCC voided Slab at KM. 17.472	RCC voided Slab at KM. 17.823	RCC Slab at KM 83.285	Total quantity	Item No. in Rate Analysis	Remarks
1	2	3	4	5	6	7	8	9	10	11	12	13
		BILL NO. 6: NEW BRIDGES										
6.01	6.01	Excavation for Foundation of Structure, as per drawings and Technical Specification. in										
	a	All types of soils and Ordinary Rocks as per Technical specification Clause 304.	cum	777	858	556	1386	1368	1574	7527	(3.32+3.33)/2	
	b	Hard Rock (requiring blasting) as per Technical Specification Clause 302 and 304.	cum	0						0	3.34	
	c	Hard Rock (using controlled blasting) as per Technical Specification Clause 302 and 304.	cum	0						0	3.9	
	d	Hard Rock (blasting prohibited) as per Technical Specification Clause 304	cum	333	372	367	594	586	674	3226	3.8 A	
6.02	6.02	PCC (M 15 Grade) in Levelling course below Foundation, Pile Cap, Approach slab etc. complete as per drawings and Technical Specification Section 1700, 2100, 2500 and 2700.	cum	51	54	71	66	67	69	377	14.10	
6.03	6.03	RCC / PCC in open foundations excluding reinforcement complete as per drawings		0						0		
	a	M 30 Grade	cum	0						0	12.8 G case - I	RCC rates are taken
	b	M 35 Grade	cum	219	236	369	520	501	462	2557	12.8 H case - I	RCC rates are taken
6.04	6.04	RCC / PCC in Substructure, Wing Wall, retaining wall, Box Cell Structures, Seismic restrainer etc. excluding reinforcement, in following grades, complete as per Drawings and Technical Specification Section 1700 and 2200		0						0		
	a	M 25	cum	0						0	13.5 F (p) Case-I	
	b	M 30	cum	0						0	13.5 G (p) Case-I	
	c	M 35	cum	274	293	309	459	441	562	2539	13.5 F (h) Case-I	
6.05	6.05	RCC in Superstructure excluding reinforcement complete as per drawings and Technical Specification Section 1700 and 2300.		0						0		
	a	Solid Slab (M 30 Grade)	cum	100					71	171	14.1C Case I (i) q	
	b	Deck Slab for Composite Girder (M35 Grade)	cum	0						0	14.1D Case I (ii) q	
	c	Voided Slab (M35 Grade)	cum	0	160	160	160	160		640	14.1D Case I (i) q	
6.06	6.06	Steel Reinforcement Fe 500D in Foundation, Substructures Superstructure etc. complete as per drawings and Technical Specification Section 1600 .	MT	79	119	192	181	175	135	881	(12.40+13.6+14.2)/3	Average of three
6.07	6.07	Steel Girder for Steel Composite Superstructure including railing and fixing of girder with Bearing complete as per drawings and Technical Specification 1000 and 1900.	MT	0						0	14.25(i)	
6.08	6.08	Bituminous (Type 2) Wearing Coat as per drawings and Technical Specification Section 2700.	sqm	145	179	179	183	179	128	990	((5.8*.040) for grading I)+5.14)	
6.09	6.09	40 thk. PCC (M25) finished with 15 thk plaster (1:3) complete as per drawings and Technical Specification.	cum	2	3	3	3	3	2	17	13.5 F (p) Case-I	
6.10	6.10	Bearings, of following Type, as per drawings and Technical Specification Section 2000		0						0		
	a	Tar Paper Bearings	sqm	27					26	53	As per Market	
	b	Elastomeric Bearings	cucm	0						0	13.14	
	c	POT cum PTEE Bearings		0						0		
	i	Free / Guided Bearings, having Capacity of		0						0		
		575 T ± 25 T	nos	0						0	As per Market	
		525 T ± 25 T	nos	0						0	As per Market	
		325 T ± 25 T	nos	0						0	As per Market	
		275 T ± 25 T	nos	0	4	4	4	4		16	As per Market	
	ii	Pin Bearings, having Capacity of		0						0		
		50 T ± 12.5 T	nos	0						0	As per Market	
		75 T ± 12.5 T	nos	0						0	As per Market	
		100 T ± 12.5 T	nos	0						0	As per Market	
		125 T ± 12.5 T	nos	0						0	As per Market	
		150 T ± 12.5 T	nos	0	1	1	1	1		4	As per Market	
	iii	Metallic Guide Bearings, having Capacity of		0						0		

QUANTITY CALCULATION FOR BRIDGE AT KM - 7.490 (C-Z Road)

Abutment

No. of Abutment = 2



No.	Item	Calculation		Quantity	Total Quantity	Unit
1.	**Earthwork in excavation for foundation for structures in all types of soil excluding rock :**					
	Abutment		=	409.220 m^3	818.44	m^3
2.	**Filling loose soil in foundations & plinth, including consolidation by watering & ramming in 15cm layers**					
	Abutment		=	0.000 m^3	0	m^3
	Backfilling behind abutments:			475.218 m^3	950.44	m^3
3.	**Structural cement concrete for Plain/ Reinforced Concrete in pile cap/open foundation: R.C.C M35 :**					
	Abutment	7.7 x 13.8 x 0.8	=	85.0 m^3	170.016	m^3
4.	**Structural cement concrete for Plain/ Reinforced Concrete in substructure, pier, piercap, dirt wall, pedestals, abutment, abutmentcap, retaining wall , return wall, staircase, etc.**		Total :	112.452 m^3	224.90	m^3
	RCC M35 in abutment:-					
	Dirt Wall		=	3.308 m^3		
	Abutment Cap		=	13.230 m^3		
	Abutment	(798.2-792.3-0.7) x 1 x 13	=	67.600 m^3		
	Return wall			28.314 m^3		
5.	**Plain cement concrete M 15 leveling course below foundation below approach slabs for bridges :**					
	Below foundation :	7.9 x 14 x 0.15	=	11.060 m^3		
	Below approach :	3.5 x 0.15 x 13.5	=	7.088 m^3		
			Total :	18.148 m^3	36.295	m^3
6.	**Structural concrete grade M 30 in Reinforced Cement Concrete in approach slabs excluding reinforcement :**					
		3.5 x 13.5 x 0.35	=	16.54 m^3	33.075	m^3
7.	**Providing & fixing in position HYSD reinforcement bars**					
	a)- Foundation/pile cap,pile,dirt wall & restrainer		=	10.891 T		
	b)- Abutment and Return Wall		=	9.591 T		
	c)- Abutment Cap		=	7.938 T		
	d)- Approach slab		=	0.000 T		
		TOTAL	=	28.420 T		
		Say :		28.42 T	56.84	T
9.	**Weep Holes :**				255.6	Nos
10.	**Filter media behind abutment**				107	m^3

CALCULATION OF QUANTITY FOR RETAINING WALL

Total length of Retaining Wall =	20	m
Overall height of retaining wall =	5.4	m
Width of Heel Slab =	3.2	m
Width of Toe Slab =	0.6	m
Thickness of Stem at bottom =	0.75	m
Total width of foundation =	4.55	m
Thickness of foundation near Stem =	0.7	m
Thickness of Heel slab Tip =	0.3	m
Thickness of Toe slab Tip =	0.3	m
Thickness of Stem at top =	0.3	m
Height of Stem =	4.7	m
Thickness of filter media behind retaining wall =	0.6	m
Thickness of levelling course below foundation =	0.15	m
Angle of internal friction for Backfill Material =	32	deg
Average height of Ground Level =	3	m
C/s area of Stem =	2.4675	m^2
C/s area of Foundation =	2.425	m^2
C/s area of filter media behind retaining wall =	2.82	m^2
C/s area of backfilling material=	21.162	m^2
C/s area of Earthwork in Excavation =	14.550	
Volume of concrete in Retaining Wall Stem =	49.4	m^3
Volume of concrete in Retaining Wall Foundation =	48.5	m^3
Total Volume of Concrete in Retaining Wall =	97.9	m^3
Volume of filter media behind retaining wall =	56	m^3
Volume of backfilling material behind wall =	423	m^3
Earth work in filling in front of abutment	36	m^3
Volume of levelling course below foundation =	14.8	m^3
Volume of Earthwork in Excavation =	291.0	m^3
Total number of weep holes =	120	Nos
Total HYSD bar for retaining wall @100=	10.8	Mton
Boulder pitching in front of retaining wall =	12	m^3

SUMMARY OF QUANTITY CALCULATION FOR BRIDGE AT KM - 7.490

6.01	Earthwork in excavation for Soil in ordinary soil		777	m^3
	Earthwork in excavation for Soil in hard rock (no blasting)		333	m^3
6.02	PCC Levelling Course (M15)		51	m^3
6.03	RCC in Open Foundation =		219	m^3
6.04	RCC in Substructure (M35)=		274	m^3
6.05	RCC in Superstructure (M35)=		100	m^3
	HYSD bar reinforcement complete as per drawings and technical specifications:			
6.06		Abutment =	56.8	MTon
		Pier =	0.0	MTon
		Retaining wall =	10.8	MTon
		Crash Barrier =	1.2	MTon
		Superstructure =	10.0	MTon
		Total =	78.8	MTon

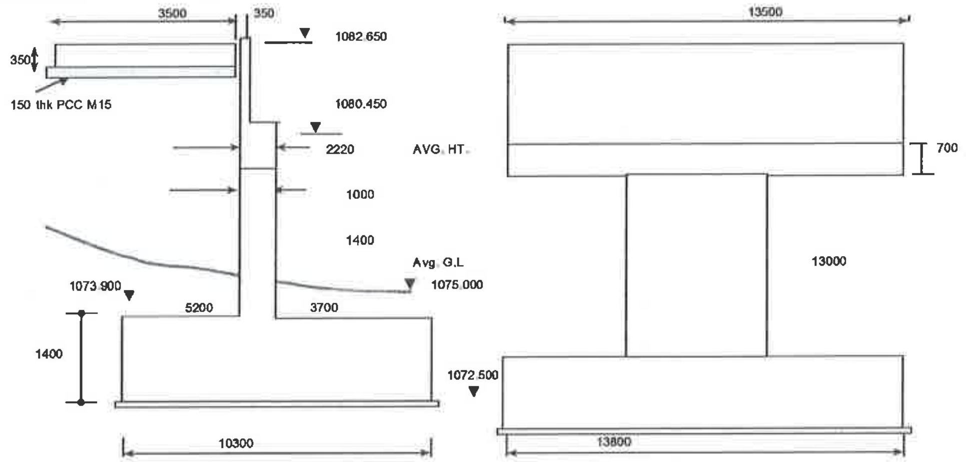
6.08	65 mm thick wearing coat	=	145	sqm
6.09	Wearing Coat on Footpath		2	m^3
6.1	Tar Paper Bearing		27	sqm
6.12	Expansion joint	Filler Type	27	m
6.13	Drainage Spout		13	Nos
6.14	Approach Slab		33	m^3
6.15	Hand Railing at the edge of Footpath		34	m
6.16	Crash Barrier		11.9	m^3
6.17	Earth work in filling		0	m
6.18	Weep holes		375.6	Nos.
6.19	Backfilling behind Abutment		1374	m^3
6.20	Filter media behind abutment		163	m^3
6.21	Boulder Grouted in Annular Space		22	m^3
6.22	M15 Grouting in Annular Space		33	m^3
6.23	Preparation of rock foundation		202.5	m^2
6.24	Lettering and No. of Bridges		100	Nos
6.25	Providing and applying 2 coats of water based cement paint		340	m^2

QUANTITY CALCULATION FOR BRIDGE AT KM -17.823 (C-Z Road)

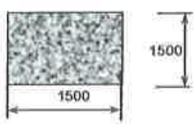
	Total Quantity	Unit

Abutment

No. of Abutment = 2



					Total Quantity	Unit
1.	**Earthwork in excavation for foundation for structures in all types of soil excluding rock :**					
	Abutment		=	746.025 m^3	1492.05	m^3
2.	**Filling loose soil in foundations & plinth, including consolidation by watering & ramming in 15cm layers**					
	Abutment		=	0.000 m^3	0	m^3
	Backfilling behind abutments:			853.686 m^3	1707.37	m^3
3.	**Structural cement concrete for Plain/ Reinforced Concrete in pile cap/open foundation: R.C.C M35 :**					
	Abutment	10.3 x 13.8 x 1.4	=	199.0 m^3	397.992	m^3
4.	**Structural cement concrete for Plain/ Reinforced Concrete in substructure, pier, piercap, dirt wall, pedestals, abutment, abutmentcap, retaining wall , return wall, staircase, etc.**		Total :	174.934 m^3	349.87	m^3
	RCC M35 in abutment:-					
	Dirt Wall		=	10.395 m^3		
	Abutment Cap		=	20.979 m^3		
	Abutment	(1080.45-1073.9-0.7) x 1.2 x 13	=	91.260 m^3		
	Return wall			50.050 m^3		
	RCC M35 in pedestals:-					
		= 1.5 x 1.5 x 0.7 x 2	=	2.2500 m^3		



					Total Quantity	Unit
5.	**Plain cement concrete M 15 leveling course below foundation below approach slabs for bridges :**					
	Below foundation :	10.5 x 14 x 0.15	=	14.700 m^3		
	Below approach :	3.5 x 0.15 x 13.5	=	7.088 m^3		
			Total :	21.788 m^3	43.575	m^3
6.	**Structural concrete grade M 30 in Reinforced Cement Concrete in approach slabs excluding reinforcement :**					
		3.5 x 13.5 x 0.35	=	**16.54** m^3	33.075	m^3
7.	**Providing & fixing in position HYSD reinforcement bars**					
	a)- Foundation/pile cap,pile,dirt wall & restrainer		=	33.528 T		
	b)- Abutment and Return Wall		=	14.131 T		
	c)- Abutment Cap		=	12.587 T		
	d)- Approach slab		=	0.000 T		
			TOTAL =	60.246 T		
			Say :	**60.25** T	120.49	T
9.	**Weep Holes :**				286.8	Nos
10.	**Filter media behind abutment**				142	m^3

Annexure -8.01

Chainage		Length	Type of Cross Section
From	To		
81+340	81+380	40	III
81+380	95+550	14170	I
95+550	95+670	120	III
95+670	95+880	210	I
95+880	95+900	20	III
95+900	97+430	1530	I
97+430	97+700	270	II
97+700	99+350	1650	I
99+350	99+370	20	II
99+370	99+800	430	I
99+800	99+860	60	II
99+860	100+000	140	I
100+000	100+070	70	II
100+070	100+100	30	I
100+100	100+200	100	II
100+200	100+260	60	I
100+260	100+280	20	II
100+280	102+850	2570	I
102+850	102+950	100	II
102+950	103+050	100	I
103+050	103+080	30	II
103+080	107+500	4420	I
107+500	107+640	140	III
107+640	108+770	1130	I
108+770	108+810	40	III
108+810	109+480	670	I
109+480	111+630	2150	V
111+630	112+700	1070	V
112+700	112+950	250	V
112+950	113+420	470	IV
113+420	113+840	420	IV
113+840	114+250	410	IV
114+250	114+330	80	III
114+330	114+470	140	IV
114+470	115+534	1064.347	IV

Length of Typical Cross Section	
Type	**Length**
I	105355
II	815
III	1340
IV	3204
V	4620
Bridge	200
Total	**115534**

Annexure -8.01

Chainage		Length	Type of Cross Section
From	To		

Summary of Drain			
S. No	**Type**	**Unit**	**Length**
1	R R Masonry Drain	m	217700
2	RCC Covered Drain	m	12444

Water seepage Locations	
Existing Location	**New Design**
From	**From**
14100	13+460
15100	14+490
25140	23+310
48050	45+990
61850	58+960
62150	63+020
62260	63+140
68250	64+890
72300	68+670
83750	79+510
84200	79+880
85350	80+670
86100	81+400
92700	87+730
94280	89+210
98275	93+030
98880	93+620
99050	93+750
99250	93+940
99450	94+130
99580	94+270
99610	93+350
102187	96+760
111075	104+700
113200	106+620
121698	114+780

No. of Location	26
Length of Affected Area at each location (average),m	20
Height of 1st cut slope,m	6
Frequency for sub surface drainage pipe of 1 m, sqm	4
Length of sub surface drain pipe	780

Project Road length	115534
No. of Culvert	649
length per culvert	178
Length of Aggregare drain	4628

CZ Road: Summary of Quantity				
Item No.	**ITEMS**	**Qty.**	**Units**	**Remarks**
8.04.a	Total Excavation for Gabion Wall =	3,62,822.8	cum	
8.04.a.i	Ordinary Soil	2,53,976.0	cum	Considering 70% of total excavation as Ordinary Soil
8.04.a.ii	Hard Rock	1,08,846.8	cum	Considering 30% of total excavation as Hard Rock
8.04.b	Backfill for Gabion Wall =	1,64,523.2	cum	
8.05	Volume of Gabian Wall =	1,77,205.4	cum	
8.10.a	RE Wall =	200.5	sqm	
8.10.b	Total Excavation for RE Wall =	6,295.3	cum	
8.10.b.i	Ordinary Soil	4,406.7	cum	Considering 70% of total excavation as Ordinary Soil
8.10.b.ii	Hard Rock	1,888.6	cum	Considering 30% of total excavation as Hard Rock
8.10.c	Backfilling behind RE Wall=	12,993.3	cum	

Calculation For Foundation Excavation Quantity and Backfill Quantity for Different Height of Gabion Wall/ Unit Run

Gabion Ht Upto GL =	0.5 m	1.0 m	1.5 m	2.0 m	2.5 m	3.0 m	3.5 m
Base Width + Projection =	1.0 m	1.5 m	2.0 m	2.0 m	3.0 m	3.0 m	3.5 m
Foundation Depth at Valley end =	0.5 m	1.0 m	1.5 m	1.5 m	1.5 m	1.5 m	1.5 m
Hill Slope angle =	50.00	50.00	50.00	50.00	50.00	50.00	50.00
Cut Slope Angle =	60.00	60.00	60.00	60.00	60.00	60.00	60.00
Height of metal thickness =	0.500	0.500	0.500	0.500	0.500	0.500	0.500
Height of cutting area above valley end (H)=	4.923	7.936	10.950	10.950	14.770	14.770	16.680
Height of Gabbion wall=	1.0	2.0	3.0	3.5	4.0	4.5	5.0
Height of Gabion wall above valley end =	0.5	1.0	1.5	2.0	2.5	3.0	3.5
Exca.Area for GW Foundation =	0.57	1.79	3.65	3.65	5.15	5.15	5.90
Exca. Area above Gabion Wall Foundation =	3.17	8.24	15.69	15.69	28.55	28.55	36.41
Base for Extra Counted Area=	1.29	1.95	2.60	2.47	3.34	3.21	3.58
Height for Extra Counted Area =	4.92	7.44	9.95	9.45	12.77	12.27	13.68
Extra counted area (sqm)=	3.17	7.24	12.96	11.69	21.34	19.70	24.49
Total Excavated area (sqm)=	0.57	2.79	6.38	7.65	12.36	14.00	17.82
Cross Section Area (sqm) =	1.50	2.75	4.13	4.63	9.25	9.75	12.75
Back Filling Area 1(sqm) =	0.29	1.15	2.60	3.54	4.62	5.85	7.22
Back Filling Area 2 (sqm) =	0	0	0	0	0	0	0
Total Back Filling area (sqm)=	0.29	1.15	2.60	3.54	4.62	5.85	7.22

Gabion Ht Upto GL =	7.5 m	8.0 m
Base Width + Projection =	6.0 m	6.0 m
Foundation Depth at Valley end =	1.5 m	1.5 m
Hill Slope angle =	50.00	50.00
Cut Slope Angle =	60.00	60.00
Height of metal thickness =	0.500	0.500
Height of cutting area above valley end (H)=	26.231	26.231
Height of Gabbion wall=	9.0	9.5
Height of Gabion wall above valley end =	7.5	8.0
Exca.Area for GW Foundation =	9.65	9.65
Exca. Area above Gabion Wall Foundation =	90.05	90.05
Base for Extra Counted Area=	5.03	4.90
Height for Extra Counted Area =	19.23	18.73
Extra counted area (sqm)=	48.40	45.92
Total Excavated area (sqm)=	51.30	53.78
Cross Section Area (sqm) =	32.25	32.75
Back Filling Area 1(sqm) =	23.38	26.05
Back Filling Area 2 (sqm) =	12.75	15.25
Total Back Filling area (sqm)=	36.13	41.30

Summary of Gabion Wall Data			
Cl. Ht (m)	X Area (m^2)	Excavation (m^3)	Backfill (m^3)
0.0	0	0	0
0.5	1.50	0.57	0.29
1.0	2.75	2.79	1.15
1.5	4.13	6.38	2.60
2.0	4.625	7.65	3.54
2.5	9.25	12.36	4.62
3.0	9.75	14.00	5.85
3.5	12.75	17.82	7.22
4.0	13.25	19.58	9.48
4.5	16.75	24.02	11.64
5.0	17.25	25.90	14.95
5.5	21.25	30.96	17.65
6.0	21.75	32.95	21.49
6.5	26.25	38.63	24.73
7.0	26.75	40.75	29.11
7.5	32.25	51.30	36.13
8.0	32.75	53.78	41.30

CZ Road: Calculations of Quantities for Road Protection Works on Valley Side of Road

Station	Height	Length	Type of Retaining Structure	Retain Ht.	Retain Length	Culvert Position	X-Area of GW (m^2)	Surface Area of RE Wall (m^2)	Volume of GW (m^3)	RE Wall Actually Provided GW (m^2)	Addl Qty GW in place replaced RE Wall	Deducted Qty GW at Culvert location (m^3)	Exca. Qty for GW (m^3)	Backfill Qty for GW (m^3)	Exca. Qty for REW (m^3)	Backfill Qty for REW (m^3)
0.00		0.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
20.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
30.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
40.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
50.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
60.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
70.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
80.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
90.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
100.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
110.00		10.00	Para W	0.00	0.00	C	0.00	0.00	0.00			0.00	0.00	0.00	0.00	0.00
120.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
130.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
140.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
150.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
160.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
169.09		9.09	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
170.00		0.91	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
180.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
184.09		4.09	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
190.00		5.91	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
197.59		7.59	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
200.00		2.41	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
210.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
220.00		10.00	Para W	0.00	0.00	C	0.00	0.00	0.00			0.00	0.00	0.00	0.00	0.00
230.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
240.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
250.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
251.25		1.25	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
260.00		8.75	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
264.75		4.75	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
270.00		5.25	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
279.75		9.75	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
280.00		0.25	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
290.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
300.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
300.49		0.49	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
310.00		9.51	Para W	0.00	0.00	C	0.00	0.00	0.00			0.00	0.00	0.00	0.00	0.00
320.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
329.90		9.90	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
330.00		0.10	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
340.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
344.90		4.90	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
345.40		0.50	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
350.00		4.60	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
360.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
366.36		6.36	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
366.86		0.50	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
370.00		3.14	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
380.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
381.86		1.86	Para W	0.00	0.00	C	0.00	0.00	0.00			0.00	0.00	0.00	0.00	0.00
390.00		8.14	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
400.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
410.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
411.01		1.01	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
420.00		8.99	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
430.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
440.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
450.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
460.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
470.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
480.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
490.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
500.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
510.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
520.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
530.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
540.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
550.00		10.00	Para W	0.00	0.00	C	0.00	0.00	0.00			0.00	0.00	0.00	0.00	0.00
560.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
570.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
580.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
590.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
600.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
610.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
620.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
630.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
640.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
645.21		5.21	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
645.31		0.10	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
645.41		0.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
650.00		4.69	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
660.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
670.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
680.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
690.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
700.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
710.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
718.87		8.87	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
720.00		1.13	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
730.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
740.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
750.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
760.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
770.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
778.35		8.35	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
780.00		1.65	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
790.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
800.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
810.00		10.00	Para W	0.00	0.00	C	0.00	0.00	0.00			0.00	0.00	0.00	0.00	0.00
812.95		2.95	Para W	0.00	0.00	C	0.00	0.00	0.00			0.00	0.00	0.00	0.00	0.00
820.00		7.05	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
830.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
840.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
850.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
854.57		4.57	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
860.00		5.43	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
870.00		10.00	Para W	0.00	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00

Summary of Unit Rates (Revised) for Chakabama-Zunheboto Road

Item No	Description	Unit	Adopted Unit Rate as per MORTH Data Book (Rs)	Item No. in Rate Analysis
1	2	3	4	5
	BILL NO. 8: DRAINAGE AND PROTECTIVE WORKS			
8.01	Lined Random Rubble Masonry Drain on Hill Side, Embankment Toe and Hill Crest Complete as per drawings and Technical Specification Section 300 and 1400 and as directed by the Engineer.	m	4936	
8.02	Covered RCC Rectangular Drain including Reinforcement complete as per drawing and Technical Specification Sections 300, 1000, 1400,1500,1600, 1700 and as directed by Engineer	m	20878	
8.03	Subsurface Drainage complete as per drawings and Technical Specification Section 309			
a	100 mm diameter PVC Perforated Closed Jointed wrapped with Geo-synthetic as per drawings and Technical Specification	m	470	
b	Aggregate Drain below Hill Side Drain as per drawings and Technical Specification	m	417	
8.04	RCC Retaining wall, Gabion Wall			
a	Excavation for RCC retaining wall, Gabion wall as per drawings and Technical Specification, in			
i	All types of soils and Ordinary Rocks as per Technical specification Clause 304.	cum	253	(3.32+3.33)/2
ii	Hard Rock (requiring blasting) as per Technical Specification Clause 302 and 304.	cum	416	3.34
iii	Hard Rock (using controlled blasting) as per Technical Specification Clause 302 and 304.	cum	409	3.9
iv	Hard Rock (blasting prohibited) as per Technical Specification Clause 304	cum	546	3.8 A
b	Back filling behind Earth Retaining Structures Complete as per drawings and Technical Specification Clause 305.	cum	3980	13.9 A
c	Boulder / Concrete Block Pitching on Cut / Embankment Slopescomplete as per drawings and Technical Specification Section 2500.	cum	4283	15.4
8.05	Gabion Structure for Retaining Earth with segments of mechanically woven (double twist) wire crates filled with boulders including non-woven geotextile as filter media complete as per drawing and Technical Specification Section 700 and 2500	cum	6325	As per Market
8.06	Seeding and Mulching of Cut Slopes / Embankment Slopes complete as per drawings and Technical Specification Clause 308	sqm	996	3.23
8.07	Geo-synthetic Mat for Erosion Control complete as per drawings and Technical Specification clause 706	sqm	1860	As per Market
8.08	Soil Nailing complete as per drawings and Technical Specification Section 3200	m	4893	As per Market
8.09	Temporary Facing complete as per drawings and Technical Specification 3200.	sqm	3045	As per Market
8.10	Reinforced Soil Wall complete as per drawing and Technical Specification Section 3100.			
a)	Facia panels as per Technical Specification Section 3105 including soil reinforcing element, foundation pad, coping beam, all accessories, consumables and components of drainage system (filter media, drainage layer, drain pipe, catch pit etc.), including ground improvement complete.	sqm	8266	7.5 (I) A (type-5) + 7.5(ii) + 12.8 B*0.20
b)	Excavation for RE wall as per drawings and Technical Specification, in			
i	All types of soils and Ordinary Rocks as per Technical specification Clause 304.	cum	253	(3.32+3.33)/2
ii	Hard Rock (requiring blasting) as per Technical Specification Clause 302 and 304.	cum	416	3.34
iii	Hard Rock (using controlled blasting) as per Technical Specification Clause 302 and 304.	cum	409	3.9
iv	Hard Rock (blasting prohibited) as per Technical Specification Clause 304	cum	546	3.8 A
c)	Back filling behind RE Wall Complete as per drawings and Technical Specification Clause 305.	cum	3980	13.9 A
8.11	Construction of Parapet Wall in RR Masonry (CM 1:3) as per Drawings and Technical Specification 1400	cum	11867	13.4 A

Quantity Calculation for Traffic Signs

Annexure -9.01

SL.NO.	NAME	CODE	LOCATION OF CAUTIONARY SIGNS																
			0KM-1KM	1KM-2KM	2KM-3KM	3KM-4KM	4KM-5KM	5KM-6KM	6KM-7KM	7KM-8KM	8KM-9KM	9KM-10KM	10KM-11KM	11KM-12KM	12KM-13KM	13KM-14KM	14KM-15KM	KM15-KM16	KM16-KM17
1	STOP	14.01		1	2		1						1	1			1	1	
2	GIVE WAY	14.02	1																
3	GIVE WAY TO BUSES EXISTING THE BUS BAY	14.03																	
4	NO STOPPING OR STANDING	14.28																	
5	NO PARKING	14.29																	
6	MAXIMUM SPEED LIMIT	14.37	2																
7	COMPULSORY KEEP LEFT	14.48	2																
8	COMPULSORY KEEP RIGHT	14.49																	
9	COMPULSORY SOUND HORN	14.57	2	2		2	2		2	2	2						2		
10	LEFT HAND CURVE	15.01	1		1								2			1			
11	RIGHT HAND CURVE	15.02	1		1								2			1			
12	RIGHT HAIRPIN BEND	15.03					1												
13	LEFT HAIRPIN BEND	15.04					1				1	1	1	1	1				
14	RIGHT HAIRPIN BEND	15.04A																	
15	RIGHT REVERSE BEND	15.05	1	1	1	1	2	1			1	1	1	1	1	1	1	1	1
16	LEFT REVERSE BEND	15.06	1	1	1	1	2		1								1	1	
17	SERIES OF BEND	15.07	1	1			2	2	1	2									1
18	SIDE ROAD RIGHT	15.09		1	2		1						1	1			1	1	
19	SIDE ROAD LEFT	15.10		1	2		1						1	1			1	1	
20	Y-INTERSECTION LEFT	15.11																	
21	Y-INTERSECTION RIGHT	15.12																	
22	Y-INTERSECTION	15.13																	
23	X-ROAD	15.14																	
24	ROUNDABOUT	15.15																	
25	T-INTERSECTION	15.17																	
26	T-INT. MAJOR ROAD	15.18																	
27	STAGGERED INTERSECTION RIGHT	15.20																	
28	STAGGERED INTERSECTION LEFT	15.21																	
29	NARROW BRIDGE AHEAD	15.25																	
30	STEEP ASCENT	15.26								1	1		1						
31	STEEP DESENT	15.27								1	1		1						
32	PEDESTRINA CROSSING	15.33		2	2														
33	SCHOOL AHEAD	15.34																	
34	BUILT UP AREA	15.35		2	2														
35	PLAYGROUND AHEAD	15.58		2															
36	BARRIER	15.60																	
37	FALLING ROCKS	15.65																	
38	SINGLE CHEVRON (SMALL)	15.72	36	45	35	27	38	51	43	47	31	41	27	42	47	36	36	40	45
39	DOUBLE CHEVRON	15.74																	
40	OBJECT MARKER(LEFT)	15.76	8	5	6	8	8	5	3	8	4	8	4	5	4	6	6	8	5
41	OBJECT MARKER(RIGHT)	15.77	8	5	6	8	8	5	3	8	4	8	4	5	4	6	6	8	5
42	TWO WAY HAZARD MARKER	15.78																	
43	MAP TYPE ADVANCE DIRECTION SIGN	16.02	3																
44	REASSURANCE SIGN	16.05																	
45	FLAG TYPE DIRECTION SIGN	16.04		2	4		2						2	2			2	2	
46	PLACE IDENTIFICATION SIGN	16.06		2	2														
47	TRUCK LAY- BY	16.07																	
48	WEIGH BRIDGE AHEAD	16.09																	
49	GANTORY BOARD	16.10	1																
50	TOILET	17.05																	
51	FILLING STATION	17.06																	
52	HOSPITAL	17.07																	
53	POLICE STATION	17.13																	
54	PICNIC SITE	17.14																	
55	HOME ZONE	17.21																	
56	BUSTOP	17.35		2	2														
57	STATE HIGHWAY ROUTE MARKER SIGN	20.01	3																
58	NATIONAL HIGHWAY ROUTE MARKER SIGN	20.02	2	2	4		1						2	2			2	2	
	TOTAL NO OF SIGNS		**73**	**77**	**73**	**47**	**70**	**64**	**53**	**69**	**45**	**59**	**50**	**61**	**57**	**51**	**59**	**65**	**57**

Quantity Calculation for Traffic Signs

Annexure -9.01

SL.NO.	NAME	CODE	LOCATION OF CAUTIONARY SIGNS														
			KM47-KM48	KM48-KM49	KM49-KM50	KM50-KM51	KM51-KM52	KM52-KM53	KM53-KM54	KM54-KM55	KM55-KM56	KM56-KM57	KM57-KM58	KM58-KM59	KM59-KM60	KM60-KM61	KM61-KM61.7
1	STOP	14.01		1	1		3		1	1	3						
2	GIVE WAY	14.02															
3	GIVE WAY TO BUSES EXISTING THE BUS BAY	14.03															
4	NO STOPPING OR STANDING	14.28															
5	NO PARKING	14.29															
6	MAXIMUM SPEED LIMIT	14.37															
7	COMPULSORY KEEP LEFT	14.48															
8	COMPULSORY KEEP RIGHT	14.49															
9	COMPULSORY SOUND HORN	14.57	4							2	4	4	4	4	2	2	
10	LEFT HAND CURVE	15.01						1									
11	RIGHT HAND CURVE	15.02						1									
12	RIGHT HAIRPIN BEND	15.03															
13	LEFT HAIRPIN BEND	15.04															
14	RIGHT HAIRPIN BEND	15.04A															
15	RIGHT REVERSE BEND	15.05	2		1	1	1			2	3	1	3	1	1	2	1
16	LEFT REVERSE BEND	15.06	2	2	1	1	1			2	3	1	3	1	1	2	1
17	SERIES OF BEND	15.07	4							1	2	2	4	4			1
18	SIDE ROAD RIGHT	15.09		1	1		3		1	1	3						
19	SIDE ROAD LEFT	15.10		1	1		3		1	1	3						
20	Y-INTERSECTION LEFT	15.11									1						
21	Y-INTERSECTION RIGHT	15.12									1						
22	Y-INTERSECTION	15.13															
23	X-ROAD	15.14															
24	ROUNDABOUT	15.15															
25	T-INTERSECTION	15.17															
26	T-INT. MAJOR ROAD	15.18															
27	STAGGERED INTERSECTION RIGHT	15.20															
28	STAGGERED INTERSECTION LEFT	15.21															
29	NARROW BRIDGE AHEAD	15.25															
30	STEEP ASCENT	15.26			1			1									
31	STEEP DESENT	15.27			1			1									
32	PEDESTRINA CROSSING	15.33		3			6				2						
33	SCHOOL AHEAD	15.34															
34	BUILT UP AREA	15.35		2			2			1	1						
35	PLAYGROUND AHEAD	15.58															
36	BARRIER	15.60															
37	FALLING ROCKS	15.65															
38	SINGLE CHEVRON (SMALL)	15.72	45	30	22	24	22	22	27	27	29	25	52	38	28	31	20
39	DOUBLE CHEVRON	15.74															
40	OBJECT MARKER(LEFT)	15.76	7	5	4	4	5	6	6	5	7	6	6	6	7	4	5
41	OBJECT MARKER(RIGHT)	15.77	7	5	4	4	5	6	6	5	7	6	6	6	7	4	5
42	TWO WAY HAZARD MARKER	15.78															
43	MAP TYPE ADVANCE DIRECTION SIGN	16.02															
44	REASSURANCE SIGN	16.05															
45	FLAG TYPE DIRECTION SIGN	16.04		2	2		6		2	2	6						
46	PLACE IDENTIFICATION SIGN	16.06		2			2			1	1						
47	TRUCK LAY- BY	16.07															
48	WEIGH BRIDGE AHEAD	16.09															
49	GANTORY BOARD	16.10															
50	TOILET	17.05															
51	FILLING STATION	17.06															
52	HOSPITAL	17.07															
53	POLICE STATION	17.13															
54	PICNIC SITE	17.14															
55	HOME ZONE	17.21															
56	BUSTOP	17.35		2			2			2							
57	STATE HIGHWAY ROUTE MARKER SIGN	20.01															
58	NATIONAL HIGHWAY ROUTE MARKER SIGN	20.02		2	2		2		2	2	2						
	TOTAL NO OF SIGNS		**71**	**58**	**41**	**34**	**63**	**38**	**46**	**55**	**78**	**45**	**78**	**60**	**46**	**45**	**33**

Annexure- 9.03

Calculation of Quantity for Overhead Gantry

S. No.	Description	Unit	No	Length (m)	Breadth (m)	Height / Depth (m)	Unit weight (Kg/m)	Quantity (Kg)
	(A) Truss							
	80mm dia NB Pipe	m	4	14.10			8.36	471.50
	40mm dia NB Pipe at Side Elevation							
	Vetical Side Pipe	m	28	1.24			3.56	123.60
	Horizontal Pipe	m	4	0.74			3.56	10.54
	Diagonal pipe at ends	m	4	1.56			3.56	22.21
	Diagonal pipe at midway	m	22	1.63			3.56	127.66
	25mm dia NB Pipe at top & bottom Elevation							
	Diagonal pipe at ends	m	4	1.22			2.41	11.76
	Diagonal pipe at midway	m	22	1.30			2.41	68.93
	Cross Pipe	m	24	0.67			2.41	38.75
	(B) Vertical Support of Pipe 300 mm Dia	m	2	5.00			49.30	493.00
							Total wt (A+B)	**1367.96**

S. No.	Description	Unit	No	Length (m)	Breadth (m)	Height / Depth (m)	Unit weight (Kg/cum)	Quantity
	(C) Gusset Plates							
	Gusset Plate at top of Truss Frame	sqm	4	1.00	1.00	0.01	7854	376.99
	Stiffners for Gusset Plate at Top frame of Truss	Sqm	8	0.56	0.30	0.01	7854	105.56
	Frame	sqm	4	1.00	1.00	0.01	7854	376.99
	Stiffners for Gusset Plate at Base of Truss Frame	Sqm	8	0.56	0.30	0.01	7854	105.56
	M.S. Plates welded to Bolts	Sqm	16	0.10	0.10	0.01	7854	12.57
							Total wt (C)	977.67
	Total weight of One Overhead Gantry (A+B+C) in (kg)							**2345.63**

Location of Proposed Gantry Signs		
S. No.	**Location**	**No. of Gantry Signs**
1	Start of Project Road	1
2	End of Project Road	1
	Total	**2**

S. No.	Description	Unit	No. of Overhead Gantry	No	Length (m)	Breadth (m)	Height / Depth (m)	Quantity	Say
1	**Gantry Mounted Overhead Signs**	Sqm	2	6	2.30	1.30		35.88	**40**
2	**Foundation Pedestal**								
	Excavation	cum	2	2	2.40	2.40	2.10	48.38	**50**
	PCC M-15 in footing	cum	2	2	2.40	2.40	0.10	2.30	**3**
	M-25 RCC in Footing	cum	2	2	2.20	2.20	0.30	5.81	**6**
	M-25 RCC in Pedestal	cum	2	2	1.00	1.00	2.45	9.80	**10**
	Reinforcemnt @ 60kg/cum of concrete	tonne	2					1.87	**2**
3	**Overhead Gantry**	tonne	2					4.69	**5**